

Touchscreens: What's The Latest?

This article explores the different advancements in touchscreen technology. It also highlights key points that design engineers must consider when selecting touchscreen technology for their projects

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"A picture is worth a thousand words. An interface is worth a thousand pictures," once said Ben Shneiderman, professor for computer science at the Human-Computer Interaction Laboratory, University of Maryland, US. Undoubtedly, the relevance of interfaces cannot be better stated. The rise in the popularity of touch-enabled devices such as smartphones, tablet PCs and all-in-one-computers in the last five years is a proof that touchscreen interfaces have become a popular choice amidst users across different segments.

No wonder, more and more design engineers are viewing the touchscreen technology with interest and incorporating it in products and surfaces that they design.

Key competing touchscreen technologies

With a plethora of touchscreen types available in the market, making the right selection is not an easy task. It is important to first understand the key technologies that comprise the touchscreen landscape and analyse how these fare against each other.

Projected-capacitive versus analogue-resistive. Projected-capacitive

and analogue-resistive are the most prevalent touch technologies.

"For more than 25 years, analogue-resistive has been the dominant technology. It is still very popular because it provides a low-cost solution, accepts input from just about any input device (pressure-sensitive) such as a credit card, pen or thick work glove, and consumes very little power," affirms Jamie D. Sewell, public relations and communications manager, Touch International.

Low cost is a key selling feature of resistive over projected-capacitive. While projected-capacitive technology can be tuned to recognise touches from various objects, it does not offer the unrestricted input options of resistive. Projected-capacitive technology was first popularised by Apple iPhone and is now the dominant new technology because it delivers beautiful optics, works with a flush-cover glass design, can potentially last forever and has multi-touch capabilities.

"As materials and production techniques continue to develop, projected-capacitive will be scaled to larger sizes and claim more market share. Going forward, we will see more penetration of this technology into medical, military and industrial applications. 3D touch (the z-axis) will also see growth in the coming years. For many applications, especially consumer products, projected-capacitive is an ideal choice. Many of Touch International's specialty customers in aerospace, medical and transportation industries are also making the switch to projected-capacitive," reveals Sewell.

Surface-acoustic wave and bending wave. Neither surface-acoustic wave (SAW) nor bending wave is truly multi-touch, which is a limiting factor for both these technologies—considering that multi-touch is all but expected in today's

